Exercise Sheet

Lecture 3

Linear Maps: Kernel, Image, Rank–Nullity

Throughout, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, and $\mathcal{L}(V,W)$ denotes the linear maps $V \to W$. Tags indicate the flavour of each problem; a ⋆ marks a harder one.

Core tutorial route

Work **3.1, 3.4, 3.8, 3.9, 3.15, 3.19, 3.20, and 3.24** in that order (about 65–80 minutes before discussion). This eight-problem route checks linearity and basis-defined maps, a complete kernel/image calculation, an image proof, rank–nullity and dimension obstructions, projection structure, and the finite injective–surjective transfer.

Additional practice: 3.2, 3.5, 3.7, 3.10–3.12, 3.14, 3.17–3.18, 3.21–3.23, 3.26. **Bridge:** 3.3, 3.6, 3.13, 3.16 (calculus applications). **Extension:** 3.25 and 3.27–3.34 (quotients and the First Isomorphism Theorem).

Linear maps

Exercise 3.1 (computational) Which of the following maps $\mathbb{R}^2 \to \mathbb{R}^2$ (or $\mathbb{R}^2 \to \mathbb{R}$) are linear?

$$T(x,y) = (x-y,\ 2x), \qquad S(x,y) = (xy,\ x), \qquad R(x,y) = |x| + y.$$

Answer. Only T is linear; S and R are not.

Exercise 3.2 (computational) Define $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ by $T(1,0) = (1,2,3)$ and $T(0,1) = (0,1,-1)$. Find a formula for $T(x,y)$, and determine $\operatorname{rank} T$ and $\dim \operatorname{null} T$.

Answer. $T(x,y) = (x,\ 2x+y,\ 3x-y)$; $\operatorname{rank} T = 2$, $\dim \operatorname{null} T = 0$.

Exercise 3.3 (computational) On $\mathcal{P}(\mathbb{R})$ let D be differentiation and M multiplication by t. Compute the commutator $[D, M^2] = DM^2 - M^2D$ as an operator.

Answer. $[D, M^2] = 2M$.

Exercise 3.4 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ satisfy $T(1,1) = (3,1)$ and $T(1,-1) = (1,3)$. Find a formula for $T(x,y)$.

Answer. $T(x,y) = (2x+y,\ 2x-y)$.

Exercise 3.5 (computational) Which of the following maps $\mathbb{R}^3 \to \mathbb{R}^2$ are linear?

$$F(x,y,z) = (x+z,\ 2y), \qquad G(x,y,z) = (x,\ yz), \qquad H(x,y,z) = (x-y,\ 0).$$

Answer. F and H are linear; G is not.

Exercise 3.6 (computational) On $\mathcal{P}(\mathbb{R})$ let D be differentiation and M multiplication by t. Compute the commutator $[D, M^3] = DM^3 - M^3D$ as an operator.

Answer. $[D, M^3] = 3M^2$.

Exercise 3.7 (proof) Fix $v_0 \in V$. Prove that the evaluation map $E\colon \mathcal{L}(V,W) \to W$ defined by $E(T) = Tv_0$ is linear.

Kernel and image

Exercise 3.8 (computational) Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be $Tx = Ax$ with $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{smallmatrix}\right)$. Find bases for $\operatorname{null} T$ and $\operatorname{range} T$, and verify the rank–nullity theorem.

Answer. $\operatorname{null} T = \operatorname{span}\big((-1,1,-1)\big)$ (dim 1); $\operatorname{range} T = \operatorname{span}\big((1,0,1),(1,1,2)\big)$ (dim 2); $3 = 2+1$.

Exercise 3.9 (proof) Suppose $v_1,\dots,v_n$ is a basis of V and $T \in \mathcal{L}(V,W)$. Prove that $\operatorname{range} T = \operatorname{span}(Tv_1,\dots,Tv_n)$.

Exercise 3.10 (computational) Let $\varphi\colon \mathbb{C}^3 \to \mathbb{C}$ be $\varphi(z_1,z_2,z_3) = z_1 + 2z_2 + 3z_3$. Describe $\operatorname{null}\varphi$ and its dimension, and decide whether φ is surjective.

Answer. $\dim\operatorname{null}\varphi = 2$; φ is surjective.

Exercise 3.11 (computational) Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be $Tx = Ax$ with $A = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{smallmatrix}\right)$. Find bases for $\operatorname{null} T$ and $\operatorname{range} T$, and verify rank–nullity.

Answer. $\operatorname{null} T = \operatorname{span}\big((-1,-1,1)\big)$ (dim 1); $\operatorname{range} T = \operatorname{span}\big((1,0,1),(2,1,1)\big)$ (dim 2); $3 = 2+1$.

Exercise 3.12 (computational) Let $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ be $Tx = Ax$ with $A = \left(\begin{smallmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 3 \\ 1 & 1 & 3 & 4 \end{smallmatrix}\right)$. Find bases for $\operatorname{null} T$ and $\operatorname{range} T$.

Answer. $\operatorname{null} T = \operatorname{span}\big((-2,-1,1,0),(-1,-3,0,1)\big)$ (dim 2); $\operatorname{range} T = \operatorname{span}\big((1,0,1),(0,1,1)\big)$ (dim 2).

Exercise 3.13 (computational) Let $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathbb{R}^2$ be $Tp = \big(p(0),\ p'(0)\big)$. Find $\operatorname{null} T$ and $\operatorname{rank} T$.

Answer. $\operatorname{null} T = \operatorname{span}(t^2, t^3)$ (dim 2); $\operatorname{rank} T = 2$ (so T is surjective).

Exercise 3.14 (proof) Let $T \in \mathcal{L}(U,V)$ and $S \in \mathcal{L}(V,W)$. Prove that $\operatorname{range}(ST) \subseteq \operatorname{range} S$ and $\operatorname{null} T \subseteq \operatorname{null}(ST)$.

Rank–nullity

Exercise 3.15 (computational) Use the rank–nullity theorem to fill in each blank. (a) $T \in \mathcal{L}(\mathbb{R}^4,\mathbb{R}^3)$ with $\dim\operatorname{null} T = 2$: find $\operatorname{rank} T$. (b) $T \in \mathcal{L}(\mathbb{R}^6,\mathbb{R}^2)$ surjective: find $\dim\operatorname{null} T$. (c) $T \in \mathcal{L}(\mathcal{P}_4(\mathbb{R}),\mathbb{R}^3)$ with $\operatorname{rank} T = 3$: find $\dim\operatorname{null} T$.

Answer. (a) $\operatorname{rank} T = 2$. (b) $\dim\operatorname{null} T = 4$. (c) $\dim\operatorname{null} T = 2$.

Exercise 3.16 (computational) Let D^2 be the second-derivative operator on $\mathcal{P}_4(\mathbb{R})$. Find $\dim\operatorname{null}(D^2)$ and $\operatorname{rank}(D^2)$.

Answer. $\dim\operatorname{null}(D^2) = 2$ and $\operatorname{rank}(D^2) = 3$.

Exercise 3.17 (computational) Let $T\colon \mathbb{R}^4 \to \mathbb{R}^4$ be $Tx = Ax$ with $A = \left(\begin{smallmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$. Find $\operatorname{rank} T$ and a basis for $\operatorname{null} T$.

Answer. $\operatorname{rank} T = 2$; $\operatorname{null} T = \operatorname{span}\big((-1,1,0,0),(0,0,-1,1)\big)$ (dim 2).

Exercise 3.18 (proof) Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$ satisfies $T^2 = 0$. Prove $\operatorname{range} T \subseteq \operatorname{null} T$, and deduce $\dim\operatorname{range} T \le \frac{1}{2}\dim V$.

Hint. What does $T^2 = 0$ tell you about Tv for each v? Combine that containment with rank–nullity.

Exercise 3.19 (proof) Suppose $P \in \mathcal{L}(V)$ satisfies $P^2 = P$ (a *projection*). Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

Exercise 3.20 ($\star$) Prove that every $T \in \mathcal{L}(\mathbb{F}^5, \mathbb{F}^2)$ has $\dim \operatorname{null} T \geq 3$.

Hint. Rank–nullity, together with the largest the rank can be when the codomain is only $\mathbb{F}^2$.

Exercise 3.21 (proof) Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that $\operatorname{rank}(S + T) \leq \operatorname{rank} S + \operatorname{rank} T$.

Hint. Show $\operatorname{range}(S+T) \subseteq \operatorname{range} S + \operatorname{range} T$, then compare dimensions using $\dim(U_1 + U_2) \leq \dim U_1 + \dim U_2$.

Operators: injective versus surjective

Exercise 3.22 (computational) Is the operator $T \in \mathcal{L}(\mathbb{R}^3)$ given by $Tx = Ax$ with $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ invertible?

Answer. Yes; $\operatorname{null} T = \{0\}$, so T is invertible.

Exercise 3.23 (computational) Is the operator $T \in \mathcal{L}(\mathbb{R}^3)$ given by $Tx = Ax$ with $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$ invertible? If not, give $\dim \operatorname{null} T$.

Answer. Not invertible; $\dim \operatorname{null} T = 1$.

Exercise 3.24 (proof) Suppose V, W are finite-dimensional with $\dim V = \dim W$, and $T \in \mathcal{L}(V, W)$ is injective. Prove that T is surjective.

Exercise 3.25 ($\star$) Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$ satisfy $ST = I$. Prove that $TS = I$ as well (so a one-sided inverse of an operator is automatically two-sided).

Hint. First show T is injective: if $Tv = 0$, apply S. In finite dimensions an injective operator is invertible.

Exercise 3.26 (proof) Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$ is surjective. Prove that T is injective.

Quotient spaces *[self-study]*

Exercise 3.27 (computational) Let $U = \operatorname{span}\big((1,1,0)\big) \subseteq \mathbb{R}^3$. Describe the coset of $(2,0,1)$ modulo U, and determine $\dim(\mathbb{R}^3/U)$.

Answer. $(2,0,1) + U = \{(2+t,\ t,\ 1) : t \in \mathbb{R}\}$; $\dim(\mathbb{R}^3/U) = 2$.

Exercise 3.28 (proof) Prove that the quotient-space operations in the lecture notes are well-defined: if $v + U = v' + U$ and $w + U = w' + U$, show $(v+w) + U = (v'+w') + U$; also show that $v + U = v' + U$ implies $av + U = av' + U$ for every scalar a.

Exercise 3.29 (computational) Let $T\colon \mathbb{R}^4 \to \mathbb{R}^2$ be $T(x_1, x_2, x_3, x_4) = (x_1 + x_2,\ x_3 - x_4)$. (a) Find $\operatorname{null} T$. (b) Write the First Isomorphism Theorem for T explicitly: what is the isomorphism $\tilde{T}\colon \mathbb{R}^4/\operatorname{null} T \to \operatorname{range} T$?

Answer. (a) $\dim \operatorname{null} T = 2$, with basis $\{(-1,1,0,0),\ (0,0,1,1)\}$. (b) $\operatorname{range} T = \mathbb{R}^2$, hence $\tilde{T}\colon \mathbb{R}^4/\operatorname{null} T \xrightarrow{\cong} \mathbb{R}^2$.

Exercise 3.30 (computational) Let $U = \operatorname{span}\big((1,1,1)\big) \subseteq \mathbb{R}^3$. Describe the coset of $(2,0,1)$ modulo U, and determine $\dim(\mathbb{R}^3/U)$.

Answer. $(2,0,1) + U = \{(2+t,\ t,\ 1+t) : t \in \mathbb{R}\}$; $\dim(\mathbb{R}^3/U) = 2$.

Exercise 3.31 (computational) Let $U = \operatorname{span}\big((1,0,1,0),(0,1,0,1)\big) \subseteq \mathbb{R}^4$. Do $(1,1,1,1)$ and the zero vector lie in the same coset modulo U? Find $\dim(\mathbb{R}^4/U)$.

Answer. Yes (since $(1,1,1,1) \in U$); $\dim(\mathbb{R}^4/U) = 2$.

Exercise 3.32 (computational) Let D be differentiation on $\mathcal{P}_3(\mathbb{R})$. Identify $\operatorname{null} D$ and the isomorphism $\mathcal{P}_3(\mathbb{R})/\operatorname{null} D \cong \operatorname{range} D$ furnished by the First Isomorphism Theorem.

Answer. $\operatorname{null} D = \operatorname{span}(1)$ (the constants); $\mathcal{P}_3(\mathbb{R})/\operatorname{null} D \cong \mathcal{P}_2(\mathbb{R})$, of dimension 3.

Exercise 3.33 ($\star$) Let $T \in \mathcal{L}(V,W)$. Prove that $\tilde{T}\colon V/\operatorname{null} T \to W$, defined by $\tilde{T}(v + \operatorname{null} T) = Tv$, is well-defined, linear, and injective.

Hint. For well-definedness, replace v by another representative of its coset and check the value is unchanged; for injectivity, see what $\tilde{T}(v + \operatorname{null} T) = 0$ forces.

Exercise 3.34 (proof) Let U be a subspace of a finite-dimensional space V. Prove that the quotient map $\pi\colon V \to V/U$, $\pi(v) = v + U$, is linear and surjective with $\operatorname{null} \pi = U$, and deduce $\dim(V/U) = \dim V - \dim U$.